



Enantioselective Grignard addition to nitroolefin



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ABSTRACT

A novel synthesis of naproxen utilizing nitroaldol, 1,4-addition and Nef reaction is demonstrated where the desired *S*-enantiomer is achieved by resolution. Further to this first enantioselective 1,4-addition of Grignard reagent to nitroolefin is demonstrated to synthesize derivatives of naproxen.

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Naproxen **1** as shown in Figure 1 is a nonsteroidal anti-inflammatory drug (NSAID)¹ meant for the treatment of pain, fever, and inflammation. Although, there are myriad approaches² that have been developed for **1**, majority of them use the process shown in Scheme 1,³ that involves overall five tedious steps. In particular, synthesis of **1** starts with 2-acetyl-6-methoxy naphthalene **7**, which upon Darzen condensation in the presence of potassium secondary butoxide with isopropylester of chloroacetic acid followed by hydrolysis and decarboxylation afforded 2-(6-methoxynaphthalen-2-yl)propanal **5**. Thereafter, the condensation of **5** with hydroxyl amine in the presence of sulfuric acid followed by dehydration and hydrolysis with caustic lye afforded the *racemic* naproxen **2** via **4** and **3**.

In our endeavor, we attempted to develop an enantioselective synthesis of **1**. In this effort, we end up unearthing first enantioselective Grignard addition to nitroolefin. Interestingly, enantioselective 1,4-addition of organometallic reagents to nitroolefin or enones is indeed one of the most elegant methods for the synthesis of enantiomerically pure substituted nitroalkanes or saturated ketones/aldehydes. Nevertheless, there has been excellent research work published in this context.^{4–6}

In order to use reactive Grignard reagents in these reactions, essentially a stoichiometric amount of a chiral additive is required to obtain a significant level of enantioselectivity.⁷ Catalytic method for the asymmetric 1,2-addition of Grignard reagents has recently been reported.⁸ However to the best of our knowledge a catalytic

enantioselective Grignard reagent addition to nitroolefin in 1,4-fashion remains unprecedented.

Herein, we disclose novel and short formal synthesis for naproxen **1** featuring nitroaldol, dehydration, 1,4-addition, and Nef reaction. Further to this first enantioselective 1,4-addition of Grignard reagents to α,β -unsaturated nitro derivatives, catalyzed by a Cu(I) and chiral ligand derived species.

The synthesis commences with the reaction of 6-methoxy-2-naphthaldehyde **11** with nitromethane in the presence of base in acetic acid (in situ salt) at 90–100 °C for 10–12 h followed by distillation of solvents under reduced pressure at <80 °C and isolation from water at pH 7. Screening of the bases (in situ salts) suggests that the ammonium acetate⁹ is the choice of reagent for nitroaldol followed by dehydration to afford intermediate **10**.

The above isolated nitroolefin **10** was treated with 3 M methyl magnesium chloride in THF at 0–5 °C in toluene for about 30–60 min followed by quenching with ammonium chloride solution at 0–5 °C to obtain 1,4-addition product nitroalkane intermediate **9**.

Thereafter, nitroalkane intermediate **9** was subjected to Nef reaction conditions by employing sodium nitrite in acetic acid and DMSO for 24 h at 40–50 °C. After the completion of the reaction (TLC), the reaction mass was quenched into 10% hydrochloric acid, stirred for about 15 min and the product was extracted into toluene. The mixture of the organic layer and water, stirred about 15 min at 25–35 °C, adjusted pH 13–14 with caustic lye, separated the toluene and washed with toluene to remove the impurities followed by extraction into toluene at pH 3–4. Distillation of toluene afforded *racemic* naproxen **2** (Scheme 2).

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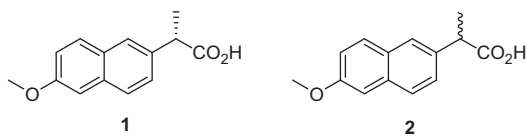
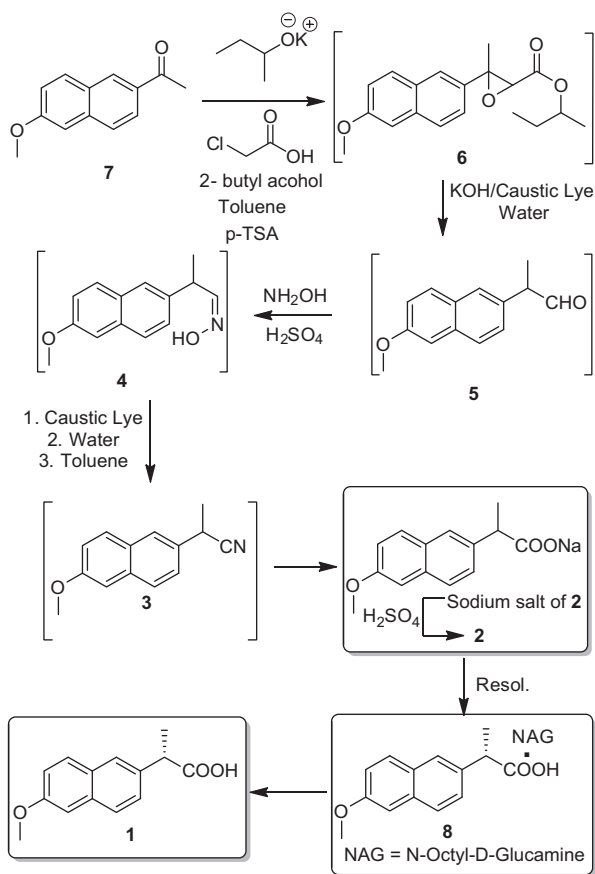


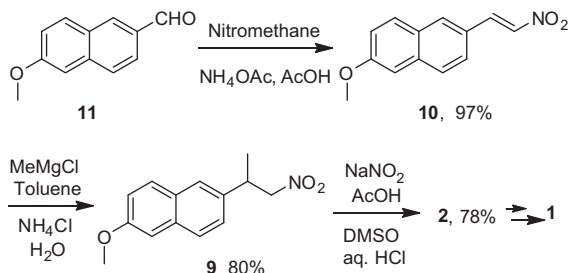
Figure 1. Naproxen frame work.



Scheme 1. Precedented synthesis for 1.

Screening of NaNO_2 equivalence at 90–100 °C suggests that the 3 equiv NaNO_2 (Table 1; entry 2) is the optimal condition for affording the racemic naproxen 2.

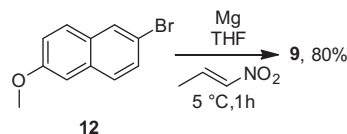
After completing the above sequence, we were intrigued by the potential of a transformation switch strategy involving the Grignard reagent of 2-methoxy-6-bromo naphthalene 12 and its 1,4-addition onto the 1-nitroprop-1-ene to afford the intermediate 9 as shown in Scheme 3.



Scheme 2. Novel synthetic approach for 1.

Table 1
Nef reaction conditions

S. no.	NaNO_2 (equiv)	Temp (°C)	Time (h)	Yield of two (%)
1	3	70	24	72
2	3	100	24	78
3	3	120	24	78
4	4	100	24	76
5	6	100	24	75



Scheme 3. Alternate approach to intermediate 9.

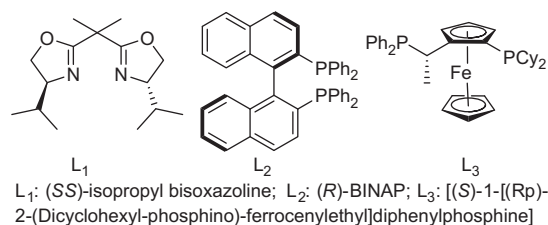
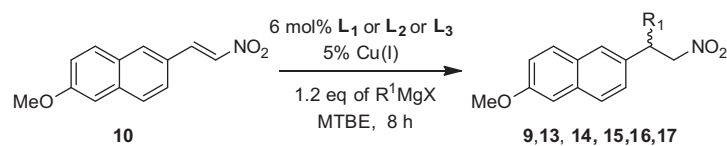


Figure 2. Structure of ligands.

In this approach, 1-nitroprop-1-ene (*precaution: unsafe to handle; dry distillation should be avoided*) was added to the Grignard reagent of 2-methoxy-6-bromo naphthalene 12 in THF at 0–5 °C for about 30–60 min that afforded 1,4-addition product, 9 in excellent yield.

Having gained intellectual control over reaction sequence, we decided to develop enantioselective version of 1,4-addition. In our endeavor, we screened various catalysts by involving ligands (L_1 – L_3) as shown in Figure 2 and Cu(I) combinations to achieve enantioselectivity.

We started the screening of the catalytic system by using 6 mol % of (*S,S*)-isopropyl bisoxazoline (L_1) and (*R*)-BINAP (L_2) ligands in combination with 5 mol % of CuTC or CuI or Zn(OTf)₂ as a metal source, 1.2 equiv of methyl magnesium halide in THF or toluene or dichloromethane and Michael acceptor 10 to obtain 9 as shown in Table 2 (entries 1 and 15). No selectivity was observed in any of the experiments conducted. Undeterred with the results, we continued our efforts to understand the ligand and catalyst quantity. In a second set of experiments, we employed 20 mol % of (*S,S*)-isopropyl bisoxazoline (L_1) and (*R*)-BINAP (L_2) ligands in combination with 15 mol % of CuTC or CuI in THF, 1.2 equiv methyl magnesium halide in THF, Michael acceptor 10 to obtain 9 as shown in Table 2 (entries 16 and 19). We did not observe any selectivity in all the experiments but we continued our efforts also to understand the steric effect of Grignard reagents and solvents. In a third set of experiments ethyl, isopropyl, and *tert*-butyl Grignard reagents were screened in *t*-BuOMe (MTBE) or THF or CH_2Cl_2 solvent by using ligand 20 mol % of (*S,S*)-isopropyl bisoxazoline L_1 and (*R*)-BINAP L_2 ligands in combination with 15 mol % of CuTC. Eventually, we observed a slight improvement in selectivity as shown in Table 2 (entries 23 and 24). Moreover, third set of experiments revealed that there is a steric effect arising from the combination of the reagent and solvent. However, there is no major impact of L_1 and L_2 ligands. Optimistically, we started exploring the possibility of improving the selectivity with different class of ligands, for example, chiral ferrocenyl diphosphine (L_3). Thus, in

Table 21,4-Addition by using ligand **L**₁, **L**₂, and **L**₃

S. no.	Catalyst	R ¹ /X	Temp (°C)	Solvent	er (R/S) (HPLC)	Yield (%)	Product
1	CuTC/L ₁	Me/Cl	−60	THF	50/50	75	9
2	CuTC/L ₁	Me/Cl	−20	THF	49/51	75	9
3	CuI/L ₁	Me/Cl	−60	THF	45/55	70	9
4	CuI/L ₁	Me/Cl	−20	THF	49/51	70	9
5	Zn(OTf) ₂ /L ₁	Me/Cl	−60	THF	50/50	70	9
6	Zn(OTf) ₂ /L ₁	Me/Cl	−20	THF	50/50	70	9
7	CuTC/L ₂	Me/Cl	−60	THF	49/51	75	9
8	CuTC/L ₂	Me/Cl	−20	THF	50/50	75	9
9	CuI/L ₂	Me/Cl	−60	THF	49/51	70	9
10	CuI/L ₂	Me/Cl	−20	THF	50/50	70	9
11	Zn(OTf) ₂ /L ₂	Me/Cl	−60	THF	49/51	70	9
12	Zn(OTf) ₂ /L ₂	Me/Cl	−20	THF	50/50	70	9
13	CuTC/L ₁	Me/Cl	−70	THF	49/51	70	9
14	CuTC/L ₁	Me/Cl	−20	DCM	49/51	70	9
15	CuTC/L ₁	Me/Cl	−35	Toluene	48/52	72	9
16	CuTC/L ₁	Me/Cl	−60	THF	48.5/51.5	72	9
17	CuTC/L ₂	Me/Cl	−60	THF	48/52	70	9
18	CuI/L ₁	Me/Cl	−60	THF	47/53	74	9
19	CuI/L ₂	Me/Cl	−60	THF	48/52	75	9
20	CuTC/L ₁	Me/Cl	−40	MTBE	48.5/51.5	70	9
21	CuTC/L ₁	Et/Cl	−40	MTBE	49/51	69	13
22	CuTC/L ₁	ⁱ Pr/Br	−40	MTBE	49/51	67	14
23	CuTC/L ₁	^t Bu/Cl	−40	MTBE	72/28	68	15
24	CuTC/L ₁	^t Bu/Cl	−70	MTBE	76.5/23.5	69	15
25	CuTC/L ₁	^t Bu/Cl	−70	THF	52/48	65	15
26	CuTC/L ₁	^t Bu/Cl	−70	DCM	53/47	66	15
27	CuTC/L ₂	^t Bu/Cl	−70	MTBE	57/43	60	15
28	CuI/L ₃	Me/Cl	−70	MTBE	49.2/50.8	68	9
29	CuI/L ₃	Et/Cl	−70	MTBE	49/51	64	13
30	CuI/L ₃	ⁱ Pr/Br	−70	MTBE	49.1/50.9	64	14
31	CuTC/L ₃	^t Bu/Cl	−70	MTBE	61/39	62	15
32	CuI/L ₃	^t Bu/Cl	−70	MTBE	98.5/1.5	62	15
33	CuTC/L ₃	Ph/Cl	−70	MTBE	50.5/49.4	62	16
34	CuTC/L ₃	Benzyl/Cl	−70	MTBE	49.4/50.5	60	17
35	CuI/L ₃	^t Bu/Cl	−70	DCM	70/30	58	15
36	CuI/L ₃	^t Bu/Cl	−70	THF	75/25	62	15
37	CuI/L ₃	^t Bu/Cl	−70	MTBE	97.9/2.1	64	15
38	CuI/L ₃	^t Bu/Cl	−70	MTBE	97.9/2.1	65	15
39	CuI/L ₃	^t Bu/Cl	−70	MTBE	97.8/2.2	63	15

HPLC: agilent, model no.: 1260, Chiralcell OJ 250 × 4.6, 5 μm, *n*-hexane, ethanol and acetic acid = 6:4:0.1/9:1:0.001, 0.9 mL/min, 240 nm.

the fourth set of experiments we involved ligand **L**₃ keeping the reaction conditions almost similar to those in the third set. Here excellent selectivity was observed for compound **15** with moderate yield as shown in Table 2 (entry 32).

The catalyst precursor of CuI along with **L**₁ or **L**₂ was experimentally compared to other commonly applied Cu(I) salt, for example, CuTC and it turned out to be superior in comparison to other salts (*vide infra*). The influence of the solvent on the selectivity of the 1,4-addition was studied with the Cu(I)/**L**₃ catalyst. This revealed that the ethereal solvents performed better in terms of stereoselectivity. Less bulky solvent furnished the 1,4-addition product with poor selectivity whereas sterically more bulky ethers, for example, MTBE provided the best enantioselectivities (Table 2; entries 23, 24, 31, and 32). Other solvents such as THF and CH₂Cl₂ led to almost *racemic* products (Table 2; entries 25, 26, 35, and 36). Eventually, MTBE as a solvent was found to be a choice for further studies. Importantly, with branched-chain and bulky Grignard reagents such as ^tBuMgCl a dramatic improvement in the enantioselectivity to an *er* of 98.5/1.5 was observed (Table 2; entry 32).

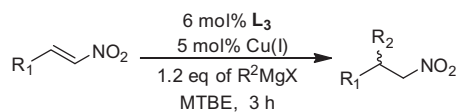
Considering the optimized conditions, establishing the reproducibility of this transformation was achieved by successfully conducting three consecutive batches (Table 2; entries 37, 38, and 39).

Substrate scope was also explored. We employed various olefins derived from benzaldehyde, 4-ethoxybenzaldehyde, 4-methoxybenzaldehyde, 4-chlorobenzaldehyde, 4-fluorobenzaldehyde, and naphthaldehyde in the 1,4-addition of ^tBuMgCl and *tert*-pentyl MgCl as Grignard reagent in MTBE. Some of the substrates have rendered excellent selectivities (Table 3; entries 3, 5, 7, and 10).

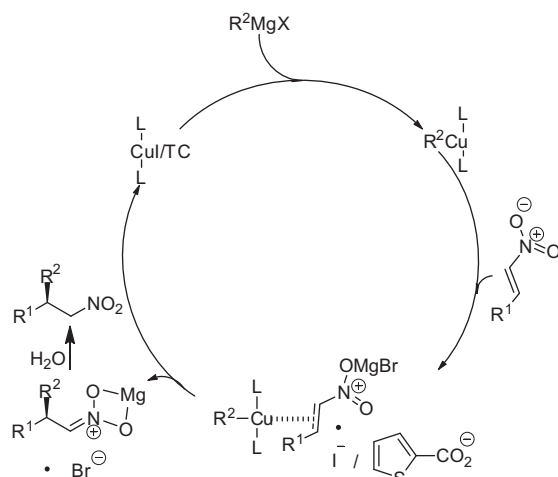
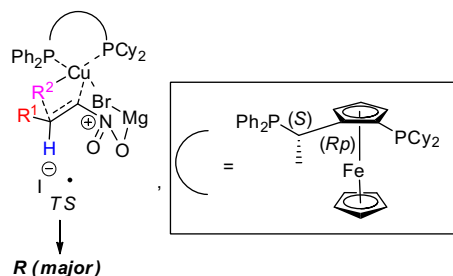
As evident in the various reactions, the chiral ferrocenyl diphosphine ligand **L**₃ afforded best enantioselectivity, which can be envisaged by considering the proposed catalytic cycle outlined in Figure 3.

The working hypothesis may involve the transfer of an alkyl ligand from the organometallic reagent to the Cu(I).

Subsequent complexation of the [−]MgBr by product with the nitro group and formation of the π -complex of the alkyl copper species with the C–C double bond of the activated olefin can result in

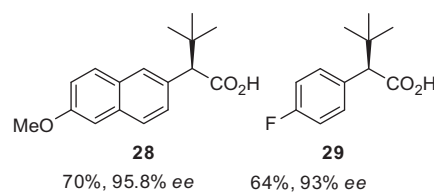
Table 3Screening of various nitroolefins and tertiary Grignard's by using ligand **L**₃ in MTBE

S. no.	Catalyst	R ¹ /R ² /X	Temp (°C)	er (R/S) (HPLC)	Yield (%)	Product
1	CuI/L ₃	Ph/ ^t Bu/Cl	−70	60/40	60	18
2	CuI/L ₃	4-EtO-Ph/ ^t Bu/Cl	−70	77/23	65	19
3	CuI/L₃	4-F-Ph/^tBu/Cl	−70	96.5/3.5	60	20
4	CuI/L ₃	4-MeO-Ph/ ^t Bu/Cl	−70	71.4/28.6	57	21
5	CuI/L₃	4-Cl-Ph/^tBu/Cl	−70	99/1	55	22
6	CuI/L ₃	Np/ ^t Bu/Cl	−70	84.3/15.7	58	23
7	CuI/L₃	6Mn/^tPentyl/Cl	−70	99/1	48	24
8	CuI/L ₃	Np/ ^t Pentyl/Cl	−70	87.8/12.2	50	25
9	CuI/L ₃	4-MeO-Ph/ ^t Pentyl/Cl	−70	73/27	52	26
10	CuI/L₃	4-F-Ph/^tPentyl/Cl	−70	96.2/3.8	45	27

HPLC: agilent, model no.: 1260, Chiralcell OJ 250 × 4.6, 5 μm, *n*-hexane, ethanol and acetic acid = 9:1:0.001, 0.9/9.5:0.5:0.001; 0.9 mL/min, 240 nm.**Figure 3.** Proposed C–C bond formation triggered by copper catalyzed conjugate addition of Grignard reagent.**Figure 4.** Proposed transition state.

the formation of the heterobimetallic complex. Subsequent alkyl transfer from the copper species to the substrate can generate the magnesium nitronate. The enantioselectivity of the overall transformation is presumed to be determined by the alkyl transfer in a preferential manner. The resulting chiral magnesium nitronate can finally decompose to afford 1,4-addition product.

Assuming the heterobimetallic species and significant steric hindrance involved in this transformation we propose a transition state where an addition of R² group occurred anti to the plane in which hydrogen is placed affording R configured product. The tran-

**Figure 5.** Structure of acid derivatives **28** and **29**.**Table 4**Conversion of nitroalkanes **15** and **20** to carboxylic acid derivatives **28** and **29** by using Nef reaction conditions

S.no.	Input	Temp (°C)	er (R/S) (HPLC)	Yield (%)	[α] _D ²⁵	Product
1	15	25	97.9/2.1	70	−77.5	28
2	20	25	96.5/3.5	64	−35	29

HPLC: agilent, model no.: 1260, (S,S) whelk-01 250 × 4.6 mm (5 μ); *n*-hexane, ethanol and acetic acid = 80:20:0.5.

sition state plausibly indicates that apart from chiral ligand (**L**₃), the size of the substituents (R¹) in nitroolefin substrate in combination with the incoming group (R²) imparts in this stereoselective transformation as shown in Figure 4.

Resulting nitroalkanes (**15** and **20**) were converted into corresponding carboxylic acids **28** and **29** by using Nef reaction conditions (Fig. 5) in good yields (70% and 64%) as well as with acceptable enantioselectivities (*ee* 95.8% and 93%) (Table 4).

In conclusion, we demonstrated the first enantioselective 1,4-addition of Grignard reagent to nitroolefin and novel *racemic* synthesis of naproxen (which was further resolved into the corresponding enantiomers) by employing nitro aldol followed by 1,4-addition and Nef reaction.

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Supplementary data

Supplementary data (experimental procedures and compound characterization data) associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.tetlet.2013.04.100>.

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